

SBA Loan Default Prediction

Clustering-Augmented Predictive Modeling

End-to-End Project Plan • 4 Juni – 17 Juni 2026 (14 Hari)

Anggota Kelompok	Algoritma yang Diimplementasi	Peran
Jason	K-Means Clustering (Unsupervised)	Tech Lead + Shared Foundation
Thariq	Logistic Regression	Predictive Model 1
Mahda	LightGBM (Gradient Boosting)	Predictive Model 2
Axel	TabPFN-3 (Foundation Model)	Predictive Model 3

Dataset	SBAnational.csv (U.S. SBA Loan, 1987–2014, ~899k baris)
Sumber	Li, Mickel & Taylor (2018), J. Statistics Education, 26(1)
Jenis Learning	Unsupervised (Clustering) + Supervised (Classification)
Target	Default (binary: 1=Charged Off, 0=Paid In Full)
Working Sample	250.000 baris (stratified by Default × Industry)
Deadline	17 Juni 2026

1. Executive Summary

Project ini bertujuan membangun pipeline machine learning untuk memprediksi probabilitas default pinjaman small business dari dataset U.S. Small Business Administration (SBA). Pendekatan utama: **clustering-augmented predictive modeling** — clustering unsupervised dipakai sebagai feature engineering, lalu hasilnya di-feed ke tiga model supervised berbeda untuk perbandingan komprehensif.

Sesuai ketentuan Final Project Pembelajaran Mesin, setiap anggota kelompok mengimplementasi **1 algoritma yang berbeda**: Jason (K-Means), Thariq (Logistic Regression), Mahda (LightGBM), dan Axel (TabPFN-3). Total 4 algoritma — 1 unsupervised + 3 supervised — saling melengkapi dalam satu pipeline end-to-end.

Key Design Decisions

- **Sampling:** Stratified subsample 250k baris dari 899k original (balance representativeness vs iteration speed).
- **Split:** 70/15/15 train/val/test, stratified by Default, seed terkunci.
- **Clustering:** K-Means (MiniBatchKMeans) fit di train only untuk hindari data leakage.
- **3 Predictive Models:** Logistic Regression (interpretable), LightGBM (strong tabular), TabPFN-3 (foundation model).
- **Evaluation:** Primary F1 (class imbalance ~18% default), secondary ROC-AUC, PR-AUC, Silhouette (clustering).
- **Skenario Uji Coba:** 3 skenario — (1) ablation with/without cluster features, (2) default vs tuned hyperparameter, (3) threshold default vs optimal.

Compliance dengan Ketentuan FP: ✓ 4 anggota, ✓ 4 algoritma berbeda, ✓ 1 dataset (SBA), ✓ Python notebook (.ipynb), ✓ Minimum 2 skenario uji coba (kita pakai 3), ✓ Format laporan BAB 1-4 (.docx), ✓ Slide presentasi (.pptx), ✓ Submission ZIP.

2. Project Context

2.1 Latar Belakang & Problem Statement

Bank yang menyalurkan pinjaman small business menghadapi keputusan kritis: apakah suatu aplikasi pinjaman harus disetujui atau ditolak? U.S. Small Business Administration (SBA) didirikan 1953 untuk mendorong pemberian pinjaman ke small business melalui loan guarantee program. Default rate yang tinggi merugikan bank, tapi penolakan berlebihan menghambat pertumbuhan ekonomi.

Dalam project ini, kami memodelkan probabilitas default sebagai binary classification problem berdasarkan fitur-fitur pinjaman (geografis, industri, tenor, jumlah, status ekonomi, dll.). Kami menambahkan clustering sebagai feature engineering — segmentasi borrower secara unsupervised — untuk melihat apakah latent structure ini meningkatkan akurasi prediksi.

2.2 Dataset Overview

Sumber data: **SBAnational.csv** dari paper Li, Mickel & Taylor (2018). Berisi 899.164 observasi dengan 27 variabel, mencakup pinjaman SBA di 50 state US dan Washington DC, periode 1987–2014, hanya untuk pinjaman dengan outcome yang sudah diketahui (paid in full atau charged off).

Variable	Type	Role dalam Project
MIS_Status	Text	Source dari target ("P I F" / "CHGOFF")
State, BankState	Text	Geografis — risk indicator
NAICS	Text	Klasifikasi industri (digunakan 2-digit prefix)
Term	Number	Tenor pinjaman (bulan) — proxy real estate backing
DisbursementGross	Currency	Jumlah pinjaman
NoEmp, CreateJob	Number	Ukuran & job creation
NewExist	Text	1=existing, 2=new business
LowDoc, RevLineCr	Text	Loan program flags
DisbursementDate	Date	Untuk derive Recession flag
GrAppv, SBA_Appv	Currency	Untuk derive Portion (SBA guarantee ratio)

Batasan Masalah (Selection Bias): Dataset hanya berisi pinjaman yang sudah disetujui dan dicairkan. Pinjaman yang ditolak di tahap awal tidak terwakili. Oleh karena itu, model yang dibangun hanya valid untuk skoring aplikasi yang mirip karakteristiknya dengan loan-loan yang historically lolos screening. Limitasi ini akan dicantumkan eksplisit di BAB 1 (Batasan Masalah) dan BAB 4 (Kesimpulan).

2.3 Target Variable Engineering

SBAnational.csv tidak punya kolom binary target — yang ada adalah MIS_Status dengan nilai text "P I F" (Paid In Full) atau "CHGOFF" (Charged Off). Target binary dibuat manual:

```
df['MIS_Status'] = df['MIS_Status'].str.strip()
df['Default'] = df['MIS_Status'].map({'CHGOFF': 1, 'P I F': 0})
df = df.dropna(subset=['Default'])
df['Default'] = df['Default'].astype(int)
```

■ **Leakage warning:** Setelah Default dibuat, drop kolom berikut sebelum training karena hanya terisi setelah outcome diketahui: MIS_Status, ChgOffDate, ChgOffPrinGr, BalanceGross.

3. Methodology Overview

3.1 Algoritma per Anggota (Compliance Check)

Sesuai aturan FP: "Setiap orang dari anggota kelompok harus mengimplementasi 1 algoritma yang berbeda." Berikut mapping algoritma:

Anggota	Algoritma	Jenis Learning	Library
Jason	K-Means (MiniBatchKMeans)	Unsupervised — Clustering	scikit-learn
Thariq	Logistic Regression	Supervised — Classification	scikit-learn
Mahda	LightGBM (Gradient Boosting)	Supervised — Classification	lightgbm
Axel	TabPFN-3 (Foundation Model)	Supervised — Classification	tabPFN

3.2 Five-Phase Approach

Phase	Owner	Output
0. Shared Foundation	Jason	Cleaned dataset, base features, encoders, split files
1. Clustering (K-Means)	Jason	Cluster features (cluster_id, distances)
2. Logistic Regression	Thariq	LogReg model + metrics (3 skenario)
3. LightGBM	Mahda	LightGBM model + metrics (3 skenario)
4. TabPFN-3	Axel	TabPFN-3 + metrics (3 skenario)
5. Comparison & Reporting	All	Ablation, laporan .docx, slide .pptx

3.3 Skenario Uji Coba (Minimum 2 — Kita Pakai 3)

Sesuai aturan FP, setiap model menjalani minimal 2 skenario uji coba. Kami menetapkan 3 skenario uji coba yang konsisten diterapkan ke ketiga supervised model (LogReg, LightGBM, TabPFN-3):

Skenario	Deskripsi	Pertanyaan yang Dijawab
Skenario 1 (Ablation Cluster Features)	Bandingkan performa model tanpa cluster features (baseline) vs dengan cluster features (enhanced).	Apakah K-Means clustering sebagai feature engineering meningkatkan akurasi prediksi?
Skenario 2 (Hyperparameter Tuning)	Bandingkan model dengan parameter default vs parameter hasil Optuna tuning (50 trials).	Seberapa besar dampak hyperparameter tuning pada performa model?
Skenario 3 (Threshold Optimization)	Bandingkan decision threshold 0.5 (default) vs threshold optimal hasil precision-recall curve.	Apakah threshold tuning di validation set meningkatkan F1 di test set?

3.4 Train/Validation/Test Split

Split	Proporsi	Jumlah Baris	Penggunaan
Train	70%	~175,000	Fit model, fit clustering, fit encoders
Validation	15%	~37,500	Hyperparameter tuning, threshold tuning
Test	15%	~37,500	Final evaluation (locked, sekali pakai)

4. Phase 0 — Shared Foundation (Jason)

Phase 0 adalah pondasi yang semua tim pakai. Tanpa ini, hasil 4 orang nggak akan comparable. Jason (tech lead) bertanggung jawab membuat semua artifact ini dan share via shared Google Drive/GitHub repo.

4.1 Task Breakdown

Task 0.1 — Data Loading & EDA

- Load SBAnational.csv. Cek dtypes, missing values, raw value distributions.
- Identifikasi kolom currency (strip '\$' dan ','): DisbursementGross, BalanceGross, ChgOffPrinGr, GrAppv, SBA_Appv.
- Identifikasi kolom date: ApprovalDate, DisbursementDate, ChgOffDate.
- Periksa MIS_Status raw values (waspadai whitespace: 'P I F' vs 'PIF').

Task 0.2 — Data Cleaning & Target Engineering

```
currency_cols = ['DisbursementGross', 'GrAppv', 'SBA_Appv',
                 'BalanceGross', 'ChgOffPrinGr']
for col in currency_cols:
    df[col] = df[col].replace(['\$', ','], '', regex=True).astype(float)

for col in ['ApprovalDate', 'DisbursementDate', 'ChgOffDate']:
    df[col] = pd.to_datetime(df[col], errors='coerce')

df.loc[~df['LowDoc'].isin(['Y', 'N']), 'LowDoc'] = pd.NA

df['MIS_Status'] = df['MIS_Status'].str.strip()
df['Default'] = df['MIS_Status'].map({'CHGOFF': 1, 'P I F': 0})
df = df.dropna(subset=['Default'])
df['Default'] = df['Default'].astype(int)
```

Task 0.3 — Feature Engineering

Replikasi derived features dari paper Li et al., plus tambahan:

Feature	Definisi
New	1 jika NewExist=2 (new business), 0 jika NewExist=1
RealEstate	1 jika Term $\geq$ 240 bulan, 0 lainnya
Recession	1 jika loan aktif Des 2007–Jun 2009, 0 lainnya
Portion	SBA_Appv / GrAppv (rasio guarantee)
Industry	NAICS 2-digit (first two characters)
SameBankState	1 jika BankState == State, 0 lainnya
LoanSize_log	$\log_{10}(\text{DisbursementGross})$
ApprovalYear	Tahun dari ApprovalDate
JobsPerLoan	$(\text{CreateJob} + \text{RetainedJob}) / \log_{10}(\text{DisbursementGross})$

Task 0.4 — Subsampling 250k & Split

```
from sklearn.model_selection import train_test_split

df['stratum'] = df['Default'].astype(str) + '_' + df['Industry'].astype(str)
counts = df['stratum'].value_counts()
valid = counts[counts >= 30].index
```

```

df_f = df[df['stratum'].isin(valid)]

df_sub, _ = train_test_split(df_f, train_size=250_000,
                             stratify=df_f['stratum'], random_state=42)
df_sub = df_sub.drop(columns=['stratum']).reset_index(drop=True)

# 70/15/15 split
leakage_cols = ['MIS_Status', 'ChgOffDate', 'ChgOffPrinGr', 'BalanceGross']
X = df_sub.drop(columns=leakage_cols + ['Default'])
y = df_sub['Default']

X_tmp, X_test, y_tmp, y_test = train_test_split(X, y, test_size=0.15,
                                                stratify=y, random_state=42)
X_train, X_val, y_train, y_val = train_test_split(X_tmp, y_tmp, test_size=0.176,
                                                  stratify=y_tmp, random_state=42)

# Save artifacts
df_sub.to_parquet('sba_250k.parquet')
np.save('train_idx.npy', X_train.index.values)
np.save('val_idx.npy', X_val.index.values)
np.save('test_idx.npy', X_test.index.values)

```

Task 0.5 — Encoding Strategy (Sepakati Bersama)

Kolom	Encoding Strategy
Industry (NAICS 2-digit)	One-hot encoding
State (51 kategori)	Target encoding via KFold (anti-leakage)
Bank (banyak unique)	Frequency encoding + group rare → "Other"
BankState	Target encoding (sama dengan State)
UrbanRural, FranchiseCode	One-hot (sedikit kategori)
LowDoc, RevLineCr	Binary mapping {Y:1, N:0}, missing→0 + flag
Numerical	StandardScaler / RobustScaler, fit di train only

Task 0.6 — Deliverables Phase 0

- File sba_250k.parquet + indices (.npy)
- Module preprocessing.py dengan fungsi get_train_val_test(include_clusters=False)
- Notebook phase0_edu.ipynb — EDA + cleaning rationale + statistik dataset
- Dokumen data_dictionary.md — penjelasan setiap fitur akhir

5. Phase 1 — K-Means Clustering (Jason)

5.1 Algoritma: MiniBatchKMeans

K-Means dipilih sebagai algoritma clustering karena: (1) scalable ke 175k train rows via MiniBatchKMeans, (2) output deterministic dan reproducible, (3) memberikan cluster_id (kategorikal) + distance ke centroid (numerikal) yang langsung bisa dipakai sebagai feature di model supervised, (4) interpretable.

5.2 Workflow

Step 1 — Feature Selection untuk Clustering

Penting: JANGAN masukkan target Default ke fitur clustering (leakage). Hindari fitur yang sangat dominan supaya cluster tidak collapse jadi split berdasarkan satu feature.

```
cluster_features = [
    'LoanSize_log', 'Term', 'NoEmp', 'CreateJob',
    'Portion', 'New', 'Recession',
    'ApprovalYear', 'JobsPerLoan',
    # + encoded Industry & target-encoded State
]
```

Step 2 — Standardize & Pilih k (Skenario Tuning)

```
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import MiniBatchKMeans
from sklearn.metrics import silhouette_score, davies_bouldin_score

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train[cluster_features])

# Cari k optimal – INI ADALAH SKENARIO TUNING UNTUK CLUSTERING
results = []
for k in range(3, 13):
    km = MiniBatchKMeans(n_clusters=k, random_state=42,
                        batch_size=4096, n_init=10)
    labels = km.fit_predict(X_train_scaled)
    idx_s = np.random.choice(len(labels), 20_000, replace=False)
    sil = silhouette_score(X_train_scaled[idx_s], labels[idx_s])
    db = davies_bouldin_score(X_train_scaled[idx_s], labels[idx_s])
    results.append({'k': k, 'inertia': km.inertia_,
                  'silhouette': sil, 'davies_bouldin': db})

# Plot elbow & silhouette → pilih k optimal
```

Step 3 — Fit Final Model & Generate Features

```
K_OPTIMAL = 7 # dari hasil step 2
km_final = MiniBatchKMeans(n_clusters=K_OPTIMAL, random_state=42,
                          batch_size=4096, n_init=20)
km_final.fit(X_train_scaled)

def add_cluster_features(X_scaled, model):
    labels = model.predict(X_scaled)
    distances = model.transform(X_scaled)
    feats = pd.DataFrame({'cluster_id': labels})
    for i in range(distances.shape[1]):
        feats[f'cluster_dist_{i}'] = distances[:, i]
    return feats

train_clust = add_cluster_features(X_train_scaled, km_final)
val_clust = add_cluster_features(X_val_scaled, km_final)
test_clust = add_cluster_features(X_test_scaled, km_final)
```

Step 4 — Cluster Interpretation (Storytelling)

- Hitung mean/distribution setiap feature per cluster.
- Hitung default rate per cluster (di training data) — **INI INSIGHT UTAMA**.
- Beri nama deskriptif, misal: "Cluster 3 = small new businesses in hospitality, high default rate (28%)".
- Visualisasi: PCA 2D scatter colored by cluster + heatmap default rate per cluster × industry.

5.3 Metric untuk Clustering (Wajib di Laporan)

- **Silhouette score** — primary metric clustering. Range [-1, 1], higher is better.
- **Davies-Bouldin index** — secondary. Lower is better.
- **Inertia (within-cluster sum of squares)** — untuk elbow plot.
- **Default rate per cluster** — business-level interpretation.

5.4 Anti-Leakage Checklist

CRITICAL:

- `scaler.fit_transform` hanya di train, `scaler.transform` di val/test.
- `km_final.fit` hanya di `X_train_scaled`.
- Cluster predictions val/test via `km_final.predict()`, bukan re-fit.
- Target Default JANGAN dimasukkan ke `cluster_features`.

5.5 Deliverables Phase 1

- File `clustering_model.pkl`, `clustering_scaler.pkl`
- File `cluster_features_{train,val,test}.parquet`
- Notebook `phase1_clustering.ipynb` — elbow/silhouette plots, profiling
- Section laporan: "Clustering Analysis"

6. Phase 2 — Logistic Regression (Thariq)

6.1 Tujuan

Logistic Regression berperan sebagai **interpretable baseline**. Koefisien LogReg dapat di-interpret sebagai odds ratio untuk feature importance business-friendly. Hasilnya juga comparable ke paper Li et al. 2018 yang juga pakai LogReg.

6.2 Implementasi 3 Skenario Uji Coba

Skenario 1 — Ablation Cluster Features:

Train LogReg (a) tanpa cluster features [baseline], (b) dengan cluster features [enhanced]. Bandingkan F1, ROC-AUC.

Skenario 2 — Hyperparameter Tuning:

Train LogReg (a) dengan default parameter ($C=1.0$, $\text{penalty}='l2'$), (b) dengan parameter hasil Optuna (50 trials di validation set).

Skenario 3 — Threshold Optimization:

Evaluate model terbaik dengan (a) threshold default 0.5, (b) threshold optimal hasil precision-recall curve di validation set.

6.3 Code Skeleton

```
from sklearn.linear_model import LogisticRegression
import optuna
from sklearn.metrics import f1_score, precision_recall_curve

# Skenario 2: Optuna tuning
def objective(trial):
    C = trial.suggest_float('C', 1e-3, 1e2, log=True)
    penalty = trial.suggest_categorical('penalty', ['l1', 'l2', 'elasticnet'])
    l1_ratio = trial.suggest_float('l1_ratio', 0, 1) if penalty=='elasticnet' else None
    model = LogisticRegression(C=C, penalty=penalty, l1_ratio=l1_ratio,
                              solver='saga', class_weight='balanced', max_iter=2000, random_state=42)
    model.fit(X_train, y_train)
    y_val_proba = model.predict_proba(X_val)[: , 1]
    return f1_score(y_val, (y_val_proba >= 0.5).astype(int))

study = optuna.create_study(direction='maximize')
study.optimize(objective, n_trials=50)

# Skenario 3: threshold tuning
y_val_proba = best_model.predict_proba(X_val)[: , 1]
prec, rec, thr = precision_recall_curve(y_val, y_val_proba)
f1s = 2*prec*rec / (prec+rec+1e-10)
best_threshold = thr[f1s.argmax()]
```

6.4 Deliverables Phase 2

- Notebook phase2_logreg.ipynb — 3 skenario terdokumentasi
- Tabel hasil: F1/ROC-AUC/PR-AUC untuk setiap skenario
- Plot: confusion matrix, ROC curve, PR curve, coefficient bar chart
- Section laporan: "Logistic Regression Results"

7. Phase 3 — LightGBM (Mahda)

7.1 Tujuan

LightGBM (Gradient Boosting Decision Trees) adalah **strong baseline industri** untuk tabular data. Bisa nangkap non-linear interactions yang nggak bisa LogReg, sekaligus memberikan SHAP-based feature importance. Diperkirakan jadi top performer di project ini.

7.2 Implementasi 3 Skenario Uji Coba (Sama dengan Phase 2)

- **Skenario 1:** Tanpa vs dengan cluster features
- **Skenario 2:** Default parameter ($n_estimators=100$, $learning_rate=0.1$) vs Optuna-tuned
- **Skenario 3:** Threshold 0.5 vs threshold optimal

7.3 Code Skeleton

```
import lightgbm as lgb
import optuna

def objective(trial):
    params = {
        'objective': 'binary', 'metric': 'binary_logloss',
        'learning_rate': trial.suggest_float('lr', 0.01, 0.3, log=True),
        'num_leaves': trial.suggest_int('num_leaves', 15, 255),
        'max_depth': trial.suggest_int('max_depth', 3, 12),
        'min_child_samples': trial.suggest_int('min_child', 5, 100),
        'subsample': trial.suggest_float('subsample', 0.5, 1.0),
        'colsample_bytree': trial.suggest_float('colsample', 0.5, 1.0),
        'reg_alpha': trial.suggest_float('reg_alpha', 1e-8, 10, log=True),
        'reg_lambda': trial.suggest_float('reg_lambda', 1e-8, 10, log=True),
        'scale_pos_weight': (y_train==0).sum() / (y_train==1).sum(),
        'random_state': 42, 'verbose': -1,
    }
    model = lgb.LGBMClassifier(**params, n_estimators=2000)
    model.fit(X_train, y_train, eval_set=[(X_val, y_val)],
              callbacks=[lgb.early_stopping(50)])
    return f1_score(y_val, (model.predict_proba(X_val)[:,-1] >= 0.5).astype(int))

study = optuna.create_study(direction='maximize')
study.optimize(objective, n_trials=50, timeout=1800)
```

7.4 SHAP Analysis (Wajib di Laporan)

```
import shap
explainer = shap.TreeExplainer(best_model)
shap_values = explainer.shap_values(X_test.sample(5000, random_state=42))
shap.summary_plot(shap_values, X_test.sample(5000, random_state=42))
```

Pertanyaan key: *Apakah cluster features muncul di top-10 importance?* Kalau iya, cluster features memang menambah signal untuk model tree-based.

7.5 Deliverables Phase 3

- Notebook phase3_lightgbm.ipynb
- Tabel hasil: 3 skenario
- Plot: confusion matrix, ROC, PR curve, SHAP summary plot, feature importance
- Section laporan: "LightGBM Results"

8. Phase 4 — TabPFN-3 (Axel)

8.1 Tujuan

TabPFN-3 adalah **foundation model untuk tabular data**, rilis Mei 2026 oleh Prior Labs. Berbeda dari model lain, TabPFN tidak melakukan gradient training — model sudah pre-trained di synthetic data, dan saat "fit" hanya menyimpan training set sebagai context untuk in-context learning. Mendukung hingga 1M training rows, jadi 175k train kita well within limit.

8.2 Setup & Workflow

```
!pip install tabpfn

from tabpfn import TabPFNClassifier
import torch

model = TabPFNClassifier(
    device='cuda',
    n_estimators=4, # smaller untuk T4 safety
    random_state=42,
)
model.fit(X_train, y_train) # no gradient, just store context
y_test_proba = model.predict_proba(X_test)[: , 1]
```

8.3 Implementasi 3 Skenario Uji Coba

- **Skenario 1:** Tanpa vs dengan cluster features
- **Skenario 2:** Default `n_estimators=4` vs tuned (eksperimen dengan `n_estimators ∈ {2, 4, 8, 16}`)
- **Skenario 3:** Threshold 0.5 vs threshold optimal

Catatan untuk Skenario 2 TabPFN: Karena TabPFN-3 minim hyperparameter (tidak ada learning rate, max_depth, dll), "tuning" di sini lebih ke eksperimen `n_estimators` sebagai ensemble size. Ini menjadikan TabPFN *simpler to tune* yang merupakan selling point — bisa di-highlight di laporan.

8.4 Backup Strategy (Kalau T4 OOM)

```
# Fallback: subsample train ke 100k
X_train_sub, _, y_train_sub, _ = train_test_split(
    X_train, y_train, train_size=100_000,
    stratify=y_train, random_state=42
)
model.fit(X_train_sub, y_train_sub)
# Test set tetap full untuk fair comparison
```

8.5 Hal Khusus untuk TabPFN-3

- **Tidak perlu heavy preprocessing** — TabPFN handle scaling internally. Tapi tetap encode kategorikal ke integer codes.
- **Feature count limit:** 200 features untuk performa optimal.
- **License:** TABPFN-3.0 License v1.0, cantumkan di laporan.
- **Runtime logging:** log wall-clock fit & predict — metric penting untuk dibandingkan.

8.6 Deliverables Phase 4

- Notebook `phase4_tabpfn.ipynb`
- Predictions: `tabpfn_test_proba_*.npy`
- Tabel hasil: 3 skenario + runtime
- Section laporan: "TabPFN-3: Foundation Model Approach"

9. Phase 5 — Comparison & Final Reporting (All Team)

9.1 Master Results Table (untuk BAB 3 Laporan)

Kumpulkan hasil semua model × semua skenario dalam satu tabel master:

Model	Skenario	F1	ROC-AUC	PR-AUC	Precision	Recall
LogReg	S1: Baseline (no cluster)	—	—	—	—	—
LogReg	S1: Enhanced (w/ cluster)	—	—	—	—	—
LogReg	S2: Default params	—	—	—	—	—
LogReg	S2: Tuned params	—	—	—	—	—
LogReg	S3: Threshold 0.5	—	—	—	—	—
LogReg	S3: Optimal threshold	—	—	—	—	—
LightGBM	(same 6 rows)	...	...	...	...	...
TabPFN-3	(same 6 rows)	...	...	...	...	...

9.2 Analisis untuk BAB 3 Pembahasan

- **Skenario 1 (Ablation):** Hitung delta F1 (enhanced – baseline) per model. Apakah konsisten positif?
- **Skenario 2 (Tuning):** Hitung improvement dari hyperparameter tuning. Model mana yang paling sensitif terhadap tuning?
- **Skenario 3 (Threshold):** Apakah threshold tuning konsisten meningkatkan F1?
- **Cross-model comparison:** Model mana yang terbaik secara F1? ROC-AUC? Runtime?
- **Statistical significance:** McNemar's test untuk klaim "Model A > Model B"

9.3 Error Analysis (untuk Diskusi)

- Identifikasi observasi yang misclassified oleh semua 3 model — "hard cases".
- Cross-tabulate misclassification rate per cluster — apakah model gagal di cluster yang sama?
- Plot calibration curve per model.

10. Timeline (4–17 Juni 2026, 14 Hari)

Timeline padat — total 14 hari dengan deadline submission 17 Juni 2026. Critical path: Phase 0 (Jason) harus selesai cepat supaya 3 anggota lain bisa mulai modeling paralel.

Hari	Tanggal	Jason	Thariq	Mahda	Axel
Hari 1	Kamis, 4 Jun	Kickoff. Mulai Phase 0: load data, EDA awal, identifikasi data issues.	Setup env, install sklearn, baca paper Li et al.	Setup env, install lightgbm, baca docs.	Setup env, install tabpfn, verify GPU.
Hari 2	Jumat, 5 Jun	Phase 0 lanjut: data cleaning, target engineering, currency/date parsing.	Bantu EDA. Mulai sketsa pipeline LogReg.	Bantu EDA. Mulai sketsa pipeline LightGBM.	Bantu EDA. Test TabPFN di toy dataset.
Hari 3	Sabtu, 6 Jun	Phase 0 finalisasi: feature engineering 8 derived features, subsampling 250k, split & encoding.	Standby — review pipeline preprocessing setelah selesai.	Standby — review pipeline.	Standby — review pipeline.
Hari 4	Minggu, 7 Jun	Phase 0 SELESAI. Distribute artifacts ke tim. Mulai Phase 1.	Mulai Phase 2: load data, run baseline LogReg (Skenario 1a).	Mulai Phase 3: load data, run baseline LightGBM (S1a).	Mulai Phase 4: load data, run baseline TabPFN (S1a).
Hari 5	Senin, 8 Jun	Phase 1 clustering: elbow plot, silhouette, pilih k.	Skenario 2: Optuna tuning LogReg.	Skenario 2: Optuna tuning LightGBM.	Skenario 2: eksperimen n_estimators TabPFN.
Hari 6	Selasa, 9 Jun	Phase 1 SELESAI. Distribute cluster features.	Lanjut tuning + analisis koefisien.	Lanjut tuning + SHAP analysis.	Lanjut eksperimen + handle T4 issues.
Hari 7	Rabu, 10 Jun	Mulai draft BAB 1 (Pendahuluan) & BAB 2 metodologi.	Skenario 1b: rerun dengan cluster features.	Skenario 1b: rerun dengan cluster features.	Skenario 1b: rerun dengan cluster features.
Hari 8	Kamis, 11 Jun	Lanjut draft laporan section clustering.	Skenario 3: threshold tuning. Eval di test set.	Skenario 3: threshold tuning. Eval di test set. SHAP final.	Skenario 3: threshold tuning. Eval di test set.
Hari 9	Jumat, 12 Jun	Phase 5: kumpulkan semua hasil, master table.	Selesaikan section laporan LogReg.	Selesaikan section laporan LightGBM.	Selesaikan section laporan TabPFN.
Hari 10	Sabtu, 13 Jun	Ablation analysis, McNemar test, error analysis.	Review section dari anggota lain.	Review section dari anggota lain.	Review section dari anggota lain.
Hari 11	Minggu, 14 Jun	Compile BAB 3 (Hasil) & BAB 4 (Kesimpulan).	Mulai bikin slide section sendiri.	Mulai bikin slide section sendiri.	Mulai bikin slide section sendiri.
Hari 12	Senin, 15 Jun	Finalisasi laporan .docx. Add flowchart desain sistem.	Slide selesai. Cleanup notebook .ipynb.	Slide selesai. Cleanup notebook .ipynb.	Slide selesai. Cleanup notebook .ipynb.
Hari 13	Selasa, 16 Jun	Final review laporan + slide. Compile semua deliverables.	Final review.	Final review.	Final review.
Hari 14	Rabu, 17 Jun	SUBMISSION DAY. Buat FP_Kelompok[X].zip. Submit.	Standby.	Standby.	Standby.

Sync Meetings

- **Daily 15-menit standup** via WhatsApp/Discord: progress, blocker.

- **Phase 0 review (akhir Hari 3, Sab 6 Jun):** Jason demo pipeline, tim approve sebelum modeling start.
- **Phase 1 review (akhir Hari 6, Sel 9 Jun):** Jason present hasil clustering & interpretation.
- **Mid-project sync (Hari 9, Jum 12 Jun):** Semua hasil eksperimen final, mulai laporan.
- **Pre-submission review (Hari 13, Sel 16 Jun):** Final check semua deliverables.

Critical Path Warning: Hari 1-3 (Phase 0) adalah bottleneck. Kalau Phase 0 mundur, semua fase berikutnya mundur. Jason harus prioritize Phase 0 above all else di 3 hari pertama. Anggota lain bisa *parallel-prep* dengan setup env, baca paper, atau eksperimen di toy dataset, tapi real modeling baru bisa start setelah Hari 4.

11. Evaluation Protocol

11.1 Metric per Algoritma (Sesuai Aturan BAB 2)

Algoritma	Primary Metric	Secondary Metrics	Visualisasi
K-Means (Jason)	Silhouette Score	Davies-Bouldin, Inertia, Default Rate/cluster	Elbow plot, PCA 2D scatter, heatmap
Logistic Regression (Thariq)	F1-Score	ROC-AUC, PR-AUC, Precision, Recall	Confusion matrix, ROC, PR, Coefficient bar
LightGBM (Mahda)	F1-Score	ROC-AUC, PR-AUC, Precision, Recall	Confusion matrix, ROC, PR, SHAP summary
TabPFN-3 (Axel)	F1-Score	ROC-AUC, PR-AUC, Precision, Recall, Confusion	Confusion matrix, ROC, PR

11.2 Mengapa F1 sebagai Primary?

Dataset punya class imbalance (~18% default). Accuracy misleading — model selalu prediksi "no default" akan dapat ~82% accuracy tapi useless. F1 balance precision dan recall, lebih sesuai untuk imbalanced binary classification.

11.3 Reproducibility Checklist

- Random seeds tercatat (numpy=42, sklearn=42, optuna=42, torch=42).
- Library versions di requirements.txt.
- Best hyperparameter tersimpan di JSON per model.
- README dengan instruksi reproduksi.

12. Deliverables & Submission Package

12.1 Format Submission (Sesuai Aturan FP)

Submission satu folder ZIP: **FP_Kelompok[X].zip** berisi 4 jenis file:

File	Format	Owner	Deskripsi
Dataset	.csv / .parquet	Jason	SBA national sumber + sba_250k.parquet processed
Code Program	.ipynb	Semua	6 notebook (1 per phase) + 1 master notebook
Dokumen Laporan	.docx	Semua	BAB 1-4 sesuai template
Slide Presentasi	.pptx	Semua	10-15 slides sesuai struktur

12.2 Struktur Repository / Folder Project

```
FP_Kelompok[X]/
├── README.md
├── requirements.txt
├── dataset/
│   ├── SBAnational.csv          # original
│   └── sba_250k.parquet         # processed subsample
├── code/
│   ├── phase0_edu.ipynb
│   ├── phase1_clustering.ipynb # Jason
│   ├── phase2_logreg.ipynb     # Thariq
│   ├── phase3_lightgbm.ipynb   # Mahda
│   ├── phase4_tabPFN.ipynb     # Axel
│   └── phase5_comparison.ipynb # All
├── laporan/
└── Laporan_FP_Kelompok[X].docx
```

```

■■■ slide/
■ ■■■ Slide_FP_Kelompok[X].pptx
■■■ results/
    ■■■ master_results.csv
    ■■■ plots/
# all evaluation plots

```

12.3 Struktur Laporan .docx (BAB 1–4)

BAB	Section	Owner	Konten Utama
BAB 1	1.1 Latar Belakang	Jason	Konteks SBA, masalah default prediction
	1.2 Rumusan Masalah	Jason	Bagaimana prediksi default? Apakah clustering membantu?
	1.3 Tujuan	Jason	Build pipeline, compare 3 model, ablation
	1.4 Manfaat	Jason	Manfaat akademik & praktis (bank, SBA)
	1.5 Batasan Masalah	Jason	Selection bias, subsampling 250k, T4 limit
BAB 2	2.1 Penjelasan Dataset	Jason	SBA description, 27 vars, contoh tampilan, target engineering
	2.2 Desain Sistem	Jason	Flowchart 5-phase + penjelasan setiap proses
	2.3 Skenario Uji Coba	All	3 skenario yang dijelaskan di Section 3.3 plan ini
BAB 3	3.1 Hasil Skenario 1 (Ablation)	All	Tabel + grafik per model
	3.2 Hasil Skenario 2 (Tuning)	All	Tabel + grafik per model
	3.3 Hasil Skenario 3 (Threshold)	All	Tabel + grafik per model
	3.4 Hasil Clustering	Jason	Silhouette, interpretasi cluster, default rate
	3.5 Pembahasan	All	Analisis & insight dari semua hasil
BAB 4	4.1 Kesimpulan	All	Jawaban rumusan masalah, key findings
	4.2 Saran	All	Future work, limitations, next steps

12.4 Struktur Slide .pptx

Slide #	Konten	Owner
1	Cover: Judul, Anggota, Mata Kuliah	All
2	Outline Presentasi	Jason
3-5	Pendahuluan: Latar Belakang, Rumusan, Tujuan, Manfaat, Batasan	Jason
6-7	Metodologi: Dataset, Desain Sistem (flowchart)	Jason
8	Skenario Uji Coba (3 skenario)	Jason
9	Hasil Clustering (K-Means)	Jason
10	Hasil Logistic Regression	Thariq
11	Hasil LightGBM	Mahda
12	Hasil TabPFN-3	Axel
13	Comparison & Ablation Analysis	All
14	Kesimpulan & Saran	All
15	Pembagian Tugas (peran setiap anggota)	Jason

13. Risk Management & Common Pitfalls

Risk	Mitigasi
Phase 0 mundur dari schedule	Jason fokus 100% di 3 hari pertama. Anggota lain standby untuk bantu kalau perlu. Buffer hari 3 untuk catch-up.
Data leakage (clustering fit di full data)	Strict rule: fit hanya di train. Code review oleh Jason sebelum cluster features didistribusikan.
Test set bocor saat tuning	Test set hanya disentuh sekali di Hari 8 (final eval). Tuning hanya pakai validation set.
Encoding inconsistency antar tim	Sepakati di Hari 3. Pakai shared preprocessing.py dengan single source of truth.
TabPFN-3 OOM di T4	Axel siap fallback subsampling 100k. Final run boleh sekali saja di T4 (jangan iterate banyak).
Cluster tidak meaningful (silhouette < 0.2)	Re-evaluate feature selection. Coba k berbeda. Kalau tetap buruk, dokumentasikan & report di laporan.
Class imbalance tidak ditangani	Konsisten pakai class_weight=balanced (LogReg) atau scale_pos_weight (LightGBM).
Optuna tuning overfit ke val	Batasi 50 trials. Validate dengan re-evaluation di val set + test set comparison.
Colab session crash / kehilangan data	Save artifacts ke Google Drive every step. Checkpoint preprocessing & model.
Anggota behind schedule	Daily standup. Jason redistribute task kalau ada blocker > 1 hari.
Deadline submission terlewat	Buffer 1 hari (16 Jun). Jangan submit di hari terakhir tanpa review.
Konflik versi notebook saat merge	Pakai Git/GitHub branches per anggota. Atau Google Drive dengan naming convention jelas.

14. References & Resources

14.1 Primary Reference

Li, M., Mickel, A., & Taylor, S. (2018). "Should This Loan be Approved or Denied?": A Large Dataset with Class Assignment Guidelines. *Journal of Statistics Education*, 26(1), 55–66. DOI: 10.1080/10691898.2018.1434342

14.2 Key Libraries & Documentation

Library	Version	URL
scikit-learn	≥1.3	scikit-learn.org
LightGBM	≥4.0	lightgbm.readthedocs.io
TabPFN	≥3.0	github.com/PriorLabs/TabPFN
Optuna	≥3.0	optuna.org
SHAP	≥0.44	shap.readthedocs.io
category_encoders	≥2.6	contrib.scikit-learn.org/category_encoders
pandas, numpy, matplotlib	latest	—

14.3 Reading List untuk Tim

- Hollmann et al. (2025). TabPFN: Accurate Predictions on Small Data with a Tabular Foundation Model. *Nature Machine Intelligence*.
- TabPFN-3 Technical Report (May 2026). Prior Labs.
- Ke, G. et al. (2017). LightGBM: A Highly Efficient Gradient Boosting Decision Tree. *NeurIPS*.
- James et al. (2013). An Introduction to Statistical Learning.
- Hosmer & Lemeshow (2000). Applied Logistic Regression.

Final Note: Project ini ambisius dengan 4 algoritma dan 3 skenario uji coba dalam 14 hari. Kunci sukses adalah disiplin pada timeline (terutama Phase 0 di 3 hari pertama) dan komunikasi harian antar anggota. Semangat tim! ■